

LINKED LISTS & TREES Solutions

COMPUTER SCIENCE MENTORS 61A

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1 Linked Lists

For the following questions, use this dataclass-based linked list. A `LinkedList[T]` is either a `Link[T]` or the empty tuple `()`. The type variable `T` describes the element type.

```
from __future__ import annotations
from dataclasses import dataclass

type LinkedList[T] = Link[T] | tuple[()]

@dataclass
class Link[T]:
    first: T
    rest: LinkedList[T] = ()

    def __str__(self) -> str:
        return format_link(self)

def format_link[T](s: Link[T]) -> str:
    string = '(' + str(s.first)
    remaining = s.rest
    while isinstance(remaining, Link):
        string += ' ' + str(remaining.first)
        remaining = remaining.rest
    assert remaining == (), 'Not a valid linked list'
    return string + ')'
```

Construct a list with `Link(1, Link(2))`. Read its elements with `first` and `rest`. At the end, `rest` is `()`, which has no `first` or `rest` attributes. `print` uses parentheses around the elements; entering a `Link` at the prompt shows the dataclass representation with field names.

1. Write `append`, which takes two linked lists and returns a new linked list with all of the elements of the first followed by all of the elements of the second. Then use `append` to write `reverse`, which returns a new linked list with the elements in reverse order.

```
def append[T](a: LinkedList[T], b: LinkedList[T]) -> LinkedList[T]:  
    """
```

```
    >>> print(append(Link(1, Link(2)), Link(7, Link(8))))
```

```
    (1 2 7 8)
```

```
    >>> print(append((), Link(5)))
```

```
    (5)
```

```
    >>> append((), ())
```

```
    ()
```

```
    """
```

```
    if _____:
```

```
        return _____
```

```
    return _____
```

```
def reverse[T](lnk: LinkedList[T]) -> LinkedList[T]:
```

```
    """
```

```
    >>> print(reverse(Link(1, Link(2, Link(3)))))
```

```
    (3 2 1)
```

```
    >>> reverse(())
```

```
    ()
```

```
    """
```

```
    if _____:
```

```
        return _____
```

```
    return _____
```

```
def append[T](a: LinkedList[T], b: LinkedList[T]) -> LinkedList[T]:
```

```
    if not isinstance(a, Link):
```

```
        return b
```

```
    return Link(a.first, append(a.rest, b))
```

```
def reverse[T](lnk: LinkedList[T]) -> LinkedList[T]:
```

```
    if not isinstance(lnk, Link):
```

```
        return ()
```

```
    return append(reverse(lnk.rest), Link(lnk.first))
```

2. Write `keep_if`, which takes a linked list of integers `lnk` and a predicate `pred`, and returns a new linked list of only the elements for which `pred` returns true, in their original order.

```
from collections.abc import Callable
def keep_if(
    lnk: LinkedList[int], pred: Callable[[int], bool]
) -> LinkedList[int]:
    """
    >>> lnk = Link(1, Link(2, Link(3, Link(4))))
    >>> print(keep_if(lnk, lambda x: x % 2 == 0))
    (2 4)
    >>> print(keep_if(lnk, lambda x: x > 0))
    (1 2 3 4)
    >>> keep_if(lnk, lambda x: x > 10)
    ()
    """
    if _____:
        return _____

    rest = _____

    if _____:
        return _____

    return _____

def keep_if(
    lnk: LinkedList[int], pred: Callable[[int], bool]
) -> LinkedList[int]:
    if not isinstance(lnk, Link):
        return ()
    rest = keep_if(lnk.rest, pred)
    if pred(lnk.first):
        return Link(lnk.first, rest)
    return rest
```

3. Write `merge`, which takes two linked lists of integers, each sorted in increasing order, and returns a new sorted linked list containing all of the elements of both.

```
def merge(a: LinkedList[int], b: LinkedList[int]) -> LinkedList[int]:  
    """
```

```
    >>> a = Link(1, Link(5))  
    >>> b = Link(2, Link(3, Link(8)))  
    >>> print(merge(a, b))  
    (1 2 3 5 8)  
    >>> print(merge(a, ()))  
    (1 5)  
    >>> merge((), ())  
    ()  
    """
```

```
    if _____:  
        return _____  
  
    elif _____:  
        return _____  
  
    elif _____:  
        return _____  
  
    else:  
        return _____
```

```
def merge(a: LinkedList[int], b: LinkedList[int]) -> LinkedList[int]:  
    if not isinstance(a, Link):  
        return b  
    elif not isinstance(b, Link):  
        return a  
    elif a.first <= b.first:  
        return Link(a.first, merge(a.rest, b))  
    else:  
        return Link(b.first, merge(a, b.rest))
```

2 Trees

For the following questions, use this dataclass-based tree. A `Tree[T]` has a label of type `T` and a list of branches, each of which is also a `Tree[T]`. A tree is never empty.

```
from __future__ import annotations
from dataclasses import dataclass, field

@dataclass
class Tree[T]:
    label: T
    branches: list[Tree[T]] = field(default_factory=list) # defaults to []

    def __str__(self) -> str:
        return format_tree(self)

def is_leaf(t: Tree) -> bool:
    return not t.branches

def format_tree(t: Tree, indent: int = 0) -> str:
    s = '  ' * indent + str(t.label)
    for b in t.branches:
        s += '\n' + format_tree(b, indent + 1)
    return s
```

4. What would Python display? Run these statements in order. If a statement causes an error, identify the error and continue.

```
>>> t = Tree(5, [Tree(3, [Tree(8)]), Tree(1), Tree(7)])
>>> t.label
```

5

```
>>> t.branches[0].label
```

3

```
>>> is_leaf(t.branches[1])
```

True

```
>>> len(t.branches)
```

3

```
>>> t.branches[1].branches[0]
```

```
IndexError: list index out of range
```

```
>>> t.branches[0].branches
```

```
[Tree(label=8, branches=[])]
```

```
>>> u = Tree(2, [t.branches[0]])
```

```
>>> u.branches[0].branches[0].label
```

```
8
```

```
>>> print(u)
```

```
2
```

```
3
```

```
8
```

5. Implement `tree_height`, which takes in a tree and returns its height: the depth of the lowest leaf, where the root has depth 0.

```
def tree_height(t: Tree[int]) -> int:
```

```
    """
```

```
    >>> tree_height(Tree(1))
```

```
    0
```

```
    >>> t = Tree(1, [Tree(4), Tree(2, [Tree(3)])])
```

```
    >>> tree_height(t)
```

```
    2
```

```
    """
```

```
    if _____:
```

```
        return _____
```

```
    return 1 + _____
```

```
def tree_height(t: Tree[int]) -> int:
```

```
    if is_leaf(t):
```

```
        return 0
```

```
    return 1 + max([tree_height(b) for b in t.branches])
```

6. Write `all_paths`, which takes a tree and returns a list of all root-to-leaf paths. Each path is a linked list of the labels from the root down to a leaf.

```
def all_paths[T](t: Tree[T]) -> list[LinkedList[T]]:
    """
    >>> t = Tree(1, [Tree(2), Tree(3, [Tree(4), Tree(5)])])
    >>> for p in all_paths(t):
    ...     print(p)
    (1 2)
    (1 3 4)
    (1 3 5)
    >>> all_paths(Tree(7))
    [Link(first=7, rest=())]
    """
    if _____:

        return _____

    return _____

def all_paths[T](t: Tree[T]) -> list[LinkedList[T]]:
    if is_leaf(t):
        return [Link(t.label)]
    return [Link(t.label, p) for b in t.branches for p in all_paths(b)]
```

7. Write `count_paths`, which takes a tree `t` of integers and an integer `total`, and returns the number of root-to-leaf paths whose labels sum to `total`.

```
def count_paths(t: Tree[int], total: int) -> int:
    """
    >>> t = Tree(1, [Tree(3), Tree(3, [Tree(4), Tree(0)]), Tree(4,
    [Tree(1)])])
    >>> count_paths(t, 4)
    2
    >>> count_paths(t, 8)
    1
    >>> count_paths(t, 6)
    1
    >>> count_paths(t, 5)
    0
    """
    if _____:

        return _____

    return _____

def count_paths(t: Tree[int], total: int) -> int:
    if is_leaf(t):
        return 1 if t.label == total else 0
    return sum([count_paths(b, total - t.label) for b in t.branches])
```

